

Statistical Methods in Model Discrimination

P. M. REILLY

Department of Chemical Engineering, University of Waterloo, Waterloo, Ontario

Particularly in the chemical kinetic field but also in others in chemistry and engineering there has been much recent attention on the use of statistics in discriminating between rival models. There are two related basic problems. The first is to design experiments which will be most informative in determining which of several possible mathematical models is the "correct" one. Simple illustrations are given of the solution of this problem by different methods, such as the criteria of Roth, Box and Hill and of others, including some work done by R. S. Hawkins and the author on expected entropy change. The second problem is to analyze the data when received. A sampling of likelihood and Bayesian approaches to this problem is described with examples.

Empirical and mechanistic models

From the statistical point of view there are at least two classes of situations where it is necessary to decide which of several mathematical models is the "best" one to represent some real-world situation.

One of these occurs in the typical engineering problem where data are available connecting, say, the yield of a chemical reaction with such controllable quantities as reactant concentrations, temperature, pressure, etc. It is desired to predict the yield (dependent variable) for combinations of the controllable quantities (independent variables) which are not included in the data but are such that each of these quantities is within the range covered by the data.

This problem is called by statisticians the response surface problem. Its treatment, using empirical models, is well described in the statistical literature⁽¹⁻¹¹⁾ and particularly⁽¹²⁾. The empirical models which have been commonly used are polynomials in the independent variables but are linear in the parameters, e.g.

$$Y_i = \theta_0 + \theta_1 X_{1i} + \theta_2 X_{2i} + \dots + \theta_p X_{pi} + \theta_{11} X_{1i}^2 + \theta_{22} X_{2i}^2 + \dots + \theta_{pp} X_{pi}^2 + \theta_{111} X_{1i}^3 + \dots + \theta_{112} X_{1i} X_{2i} + \dots + \epsilon_i \dots \dots \dots (1)$$

where Y_i is the value of the dependent variable at the i^{th} measurement, X_{ji} is the value of the j^{th} independent variable at the i^{th} measurement, the θ 's are parameters to be evaluated, and ϵ_i is the measurement error associated with the i^{th} measurement. The model discrimination problem here consists of finding out how many terms are required in the model for satisfactory prediction. When the model is actually used for prediction, values for the θ 's estimated from the data are used and ϵ_i is taken to be zero.

If it is only desired to make predictions and to do that within the ranges of the independent variables covered by the data, empirical polynomials prove very easy and flexible to use and should ordinarily be chosen.

Another type of situation occurs where one or more physical mechanisms can be suggested to account for the mathematical

On a porté beaucoup d'attention récemment à l'emploi des statistiques pour différencier entre des modèles rivaux et ce, non seulement dans le domaine de la cinétique chimique, mais aussi dans d'autres en chimie et en génie. Il existe deux problèmes fondamentaux reliés l'un à l'autre: le premier consiste à concevoir des expériences qui permettent d'obtenir le plus de renseignements possible pour déterminer lequel de plusieurs modèles mathématiques est le "bon"; on fournit des exemples simples de la solution de ce problème au moyen de différentes méthodes telles que les critères de Roth, Box et Hill et d'autres, y compris un travail fait par R. S. Hawkins et l'auteur sur un changement d'entropie prévue. Le second problème consiste à analyser les résultats reçus; on décrit avec des exemples la méthode de probabilité et la méthode bayésienne pour le résoudre.

relation between the variables and it is desired to choose from these which one represents the "best" approximation to "the truth". This type of situation arises where for scientific purposes it is desired simply to investigate the mechanism of the relationship or where it is necessary to produce a mathematical model which is valid over wider ranges of the variables.

These mechanistic models are ordinarily more complicated mathematically than the simple empirical models and are usually non-linear in both the independent variables and the parameters. This is the case, for instance, with Hougen-Watson models in chemical kinetics⁽¹³⁾. This paper concerns itself with the problem of choosing between alternative mechanistic models.

A good discussion of the problem is given by Kittrell, Mezaki and Watson⁽¹⁴⁾.

"Truth" in mathematical modelling

It is recognized that except in trivial cases the real world and the mathematical world are incommensurable; no mathematical model can perfectly represent a non-trivial real world situation. The problem of discrimination among mechanistic models is then not that of choosing which one is right and which others are wrong but that of choosing which one, hopefully unique, gives predictions which in some stated sense are better than those given by the others.

The design problem and the analysis problem

It will commonly happen that the available data are not sufficient on which to choose between different models. In Figure 1, if the measurements of the dependent variables are all made at low values of the independent variable x , no reasonable amount of data will permit a choice between Model 1 and Model 2. Even if the data are taken at higher values of x , if there is large experimental error, low values of y could possibly be explained by error, rather than by a preference towards Model 1 over Model 2.

This illustrates that there are two aspects to the application of statistical methods to model discrimination problems. The first is the design problem; that is to choose the experimental

conditions in such a way that discrimination is possible. The second is the analysis problem, that of analysing the data to assess how much discrimination has been achieved.

It is clear that the design problem is the more fundamental one. If for some reason the analysis of the data is faulty it may be repeated correctly but no amount of analysis can resurrect information from data which do not contain it because of faulty design. In presentation, it is easier to discuss the analysis problem first.

The analysis of model discrimination experiments

The methods to be discussed here are intended to be applied to models which are non-linear in the parameters and are not recommended as the best for models linear in the parameters.

There are some special methods developed by D. R. Cox^(15,16) which were pioneering works in this field and are still useful, especially for distinguishing between only two rival models.

In the general case there are two approaches which may be considered particularly useful. These are the likelihood approach and the Bayesian approach. Though these methods have superficial similarities, they are fundamentally different in principle.

Likelihood discrimination

The concept of a probability function is a familiar one. Suppose $p(x,\theta)$ is a probability function, which when given values of one or more parameters θ , allows the probability of any outcome x to be calculated. For example

$$p(x,\theta) = \frac{n!}{x!(n-x)!} \theta^x (1-\theta)^{n-x} \dots\dots\dots (2)$$

is the binomial probability function and if the values $n=5$, $\theta = \frac{1}{6}$ are substituted into it it becomes

$$\frac{5!}{x!(5-x)!} \left(\frac{1}{6}\right)^x \left(\frac{5}{6}\right)^{5-x}$$

which is a function of x alone and for instance gives the probability of x three's showing in 5 rolls of a true die. In this case n is the number of trials, x the number of threes, and θ the probability of a three in one trial.

Suppose now that a die is being tested for trueness and therefore θ is unknown. If it is rolled 5 times and a three turns up twice we may substitute the available information into the probability function to obtain

$$L(\theta) = \frac{5!}{2!3!} \theta^2 (1-\theta)^3 \dots\dots\dots (3)$$

It is now written $L(\theta)$ to emphasize that it is a function of θ only and it is called a likelihood function. If a value of θ , say θ_1 , is substituted into $L(\theta)$ it gives the probability that the event which actually happened (2 threes on 5 rolls) would have if the value of θ were θ_1 . Comparing the values of the likelihood function for two different values for θ gives the relative plausibilities of these two values in the light of the data. It is convenient to compare two likelihoods by examining their ratio, and the likelihood ratio $L(\theta_1)/L(\theta_2)$, sometimes called the odds ratio, may be taken as a direct measure of the relative plausibilities of θ_1 and θ_2 in the light of the data. In the case of the likelihood function given, $L(2/5)/L(1/6) = 2.15$. This indicates that the data obtained are 2.15 times as probable if $\theta = 2/5$ as they are if $\theta = 1/6$ and the value 2.15 can be taken as the weight of the evidence against the die being true (2/5 is the value of θ which maximizes $L(\theta)$ while $\theta = 1/6$ if the die is true). This would not ordinarily be taken as strong evidence that the die is not true. A likelihood ratio of 10 is ordinarily taken as showing a real difference in plausibility while 100 denotes strong preference for the one value of the parameter over the other. References to likelihood methods in general are ^(17,18).

A slight extension of this idea is required if we are to distinguish between two or more mathematical models ⁽¹⁹⁾. Suppose we have two models which give rise to probability functions for the observations $f_1(y,\theta_1,\theta_2)$ and $f_2(y,\phi_1,\phi_2,\phi_3)$. Then after some observations y are made, the likelihood function for the first model will be a function of two parameters θ_1 and θ_2 while for the second it will be a function of the three parameters ϕ_1, ϕ_2, ϕ_3 and some way must be found to eliminate this "nuisance" dependence on the parameters. One way of finding a likelihood ratio which applies between the two models is to examine

$$\lambda = \frac{\text{Max}_{\theta_1, \theta_2} f_1(y, \theta_1, \theta_2)}{\text{Max}_{\phi_1, \phi_2, \phi_3} f_2(y, \phi_1, \phi_2, \phi_3)} \dots\dots\dots (4)$$

This is a comparison of the likelihoods of the two models at their (individual) best. It is simply a comparison of how well the two models can be made to fit the data, expressed in likelihood terms. To compare more than two models the maximum likelihood for each model may be shown and two-way comparisons made by examining the ratios. Since only ratios are relevant between likelihoods, constants which multiply all the likelihoods in a comparison set are irrelevant.

This method of eliminating nuisance parameters by maximizing with respect to them is by no means the only way of eliminating nuisance parameters in likelihood inference. It is presented here because it is convenient to use and at least when a large amount of data is available it usually gives results consistent with those given by other methods. Likelihood methods as such have a very sound basis in theory.

When the models have different numbers of parameters there are inherent difficulties with any discrimination method. Using the likelihood method described here good discrimination requires that the likelihood ratio be higher than usual if the favoured model is the one with the larger number of parameters.

Example of model discrimination by likelihoods

Consider a simple numerical example. Suppose the vector of experimental conditions ξ consists of the single independent variable x , the dependent variable is y , and the following set of measurements is available.

i	x_i	y_i
1	0	-1.290
2	1	5.318
3	2	7.049
4	3	19.886

There are three models to be considered

- Model 1. $y_i = \theta_{11}x_i + \epsilon_i$
- Model 2. $y_i = \theta_{21} + \theta_{22}x_i + \epsilon_i \dots\dots\dots (5)$
- Model 3. $y_i = \theta_{31}\exp \theta_{32}x_i + \epsilon_i$

The error ϵ_i for each model is defined to have expectation zero and previous measurements have shown the error variance, σ^2 , to be 1. The errors are assumed normally distributed and independent of each other. Then the probability density function for the observations, assuming model j is

$$f_j(y,\theta) = \frac{1}{\sigma^4(\sqrt{2\pi})^4} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^4 [y_i - E_j(y_i)]^2 \right\} \dots\dots (6)$$

- where $E_1(y_i) = \theta_{11}x_i$ for model 1
- $E_2(y_i) = \theta_{21} + \theta_{22}x_i$ for model 2
- $E_3(y)_i = \theta_{31} \exp \theta_{32}x_i$ for model 3

This density function is the product of four normal densities, one for each observation. They can be multiplied because the errors are independent.

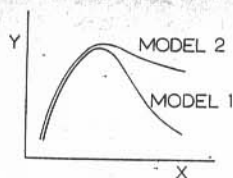


Figure 1—A comparison of two models.

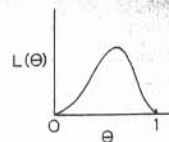


Figure 2—Likelihood function.

If the known numerical values for the x_i , y_i and σ are substituted into the above we obtain the likelihood functions

$$L_j(\theta) = \exp \left\{ -\frac{1}{2} \sum_{i=1}^4 [y_i - E_j(y_i)]^2 \right\} \quad j = 1, 2, 3 \dots (7)$$

where the constants in front of the expression have been dropped because only ratios are of concern. θ here is the parameter vector appropriate to the model.

The $L_j(\theta)$ functions are obviously maximized if we choose the θ values so as to minimize

$$Z_j = \sum_{i=1}^4 [y_i - E_j(y_i)]^2 \dots \dots \dots (8)$$

This is the least squares criterion for choosing estimates of θ . In passing it should be noted that an important part of the justification for least squares estimates is that they maximize the likelihood function if the error distribution is normal.

It follows that

$$\text{Max}_{\theta} L_j(\theta) = \exp \left\{ -\frac{1}{2} R_j \right\} \quad \text{where } R_j = \text{Min}_{\theta} Z_j \dots \dots (9)$$

R_j is the residual sum of squares after estimating the parameters of model j by least squares. In the example we have

Model (j)	R_j	$K \text{ Max}_{\theta} L_j(\theta)$
1	28.465	.050
2	22.473	1
3	11.854	202.2

The maximum likelihoods have been multiplied by a constant K in order to give them manageable values. There is seen to be good separation among the models, with model 3 preferred. Actually the data were generated artificially using model 3 with $\theta_{31} = \theta_{32} = 1$ and $\sigma = 1$.

The Bayesian approach ^(20, 21)

This is based on a subjective interpretation of probability and on Bayes' Theorem. The interpretation of probability used previously in this paper is the objective one in which the probability of an event is interpreted as its long-term relative frequency. Subjective probability is a measure of degree of belief⁽²²⁾. It is postulated that one may express his degree of belief in a proposition as a probability. We often use this interpretation of probability at least qualitatively in ordinary speech when we say "It will probably rain" (my degree of belief in rain is high) "There is a 50-50 change that I will go" (I believe equally strongly that I will go or not go).

Bayes' Theorem is a mathematical consequence of axioms which are acceptable as reasonable either with a subjective or objective interpretation of probability. It provides the method by which our opinions are updated on the basis of experimental evidence.

As an example, return to the situation where we were examining a die to see if it was true and let us assume that initially we have no idea of the value of θ , the probability of a three in one trial. We may express this by saying that our prior probability density function for θ is

$$f(\theta) = 1, 0 < \theta < 1 \\ = 0, \text{ otherwise.}$$

This is a distribution of subjective probability and expresses the same belief in every value of θ between 0 and 1, but not including them.

Bayes' Theorem states

$$Pr \{ \theta/x \} \propto Pr \{ \theta \} Pr \{ x/\theta \}$$

where $Pr \{ \theta/x \}$ is read "the probability of θ given x ", and is called the posterior probability of θ . $Pr \{ \theta \}$ is the prior probability of θ and $Pr \{ x/\theta \}$ may be recognized as the likelihood function. In our case we may use Bayes' Theorem in terms of probability density functions as

$$f_2(\theta/x) \propto f(\theta) f_1(x/\theta)$$

Then $f(\theta)$ is the prior probability density function given above and $f_1(x/\theta)$ is the likelihood function shown previously. That is

$$f_2(\theta/x) \propto 1 \times \frac{5!}{2!3!} \theta^2(1-\theta)^3, 0 < \theta < 1 \\ = 0 \text{ otherwise.}$$

or

$$f_2(\theta/x) = \frac{\theta^2(1-\theta)^3}{\int_0^1 \theta^2(1-\theta)^3 d\theta}$$

because $f_2(\theta/x)$, being a density function of θ must integrate to 1 with respect to θ . We therefore obtain

$$f_2(\theta/x) = 60 \theta^2(1-\theta)^3, 0 < \theta < 1 \dots \dots \dots (10) \\ = 0 \text{ otherwise}$$

Comparing Figure 2 and Figure 3 we see that the posterior density function for θ looks identical to the likelihood function for θ . The meaning is entirely different however. Areas under the posterior density function are subjective probabilities and are used to denote "confidence intervals" etc. The ordinates of this function have no simple meaning.

With the likelihood function of Figure 2, only the ordinates have meaning and their ratios are used to compare plausibilities, to set up plausibility intervals, etc.

Example of model discrimination by Bayes' theorem

The method outlined and illustrated here is by no means the only Bayesian one but it will show the principles.

Nuisance parameters like the parameters in the models to be examined here are handled in the Bayesian approach by "integrating them out", i.e. a result is found which is an average over all possible values of the parameters, each value weighted according to its probability. If the model is linear in the parameters and the random variables are normally distributed the equivalent of this integration may be carried out very simply.

Suppose for some particular model the prior distribution of the parameter vector θ is the multivariate normal distribution with mean θ_0 and covariance matrix V . This will be a subjective

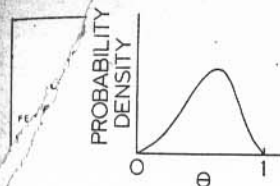
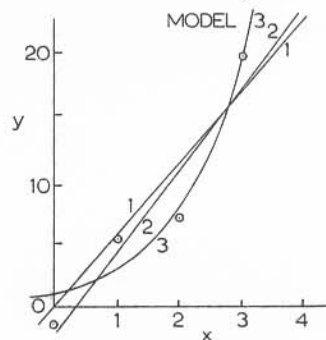


Figure 3—Posterior density function.

Figure 4—Models fitted to given data.



probability distribution which expresses whatever we know in advance about θ . Let the model be

$$y_i = f(\theta, \xi_i) + \epsilon_i$$

where y_i is the value of the i th measurement

θ is the $p \times 1$ vector of parameters

ξ_i is the $q \times 1$ vector of independent variables at the i th measurement

ϵ_i is the error associated with the i th measurement.

If there are n measurements, we may approximately linearize this model with respect to θ as follows⁽²³⁾.

$$u = X(\theta - \theta_0) + \epsilon$$

where u is an $n \times 1$ vector, the i th element of which is

$$y_i - f(\theta_0, \xi_i)$$

X is an $n \times p$ matrix, the i, j element of which is

$$\left[\frac{\partial f(\theta, \xi_i)}{\partial \theta_j} \right]_{\theta=\theta_0}$$

ϵ is an $n \times 1$ vector of errors.

Under reasonable assumptions about the mathematical nature of the model, the linearized form will be a good approximation for θ in some neighbourhood of θ_0 . The error vector is assumed to have the multivariate normal distribution with mean O and covariance matrix $I\sigma^2$ where O and I are the null and identity matrices respectively. It then follows, taking expectations and covariances on the linearized model (with respect to random variation both in θ and ϵ), that u has expectation O and covariance matrix $XVX' + I\sigma^2$. Its distribution is multivariate normal because it is a linear combination of quantities with the multivariate normal distribution. This distribution represents the total uncertainty in u arising from both uncertainty in θ and experimental error. Note that no actual integration was done here in allowing for the effect of the nuisance parameter vector θ . It was done implicitly however in taking expectations and covariances.

Applying Bayes' Theorem, -

Posterior Probability of the Model $\propto$ Prior Probability of the Model $\times g\{u \text{ given the model}\}$

where $g\{u \text{ given the model}\}$ is the multivariate normal density with mean O and covariance matrix $XVX' + I\sigma^2$ evaluated at u . ' signifies transposition.

If further data become available after these posterior model probabilities are calculated, they can be updated by a method described by Box and Hill⁽²³⁾.

To illustrate this approach we use the same example as was used for the likelihood method. We take the prior probabilities of the models to be 1/3 each showing no preference for any model and the prior distribution of θ_{11} to be $N(0,100)$, i.e. univariate normal with mean zero and variance 100, implying little knowledge of θ_{11} . Similarly we take the prior distribution of the vector of parameters of model 2 to be $N(O,100I)$ the bivariate normal distribution with mean O and covariance

matrix $100I$ where I is the 2×2 identity matrix. We take the prior distribution for the parameter vector of model 3 to be $N\left[\begin{pmatrix} 1.173 \\ .943 \end{pmatrix}, 100I\right]$. The choice of the expectation vector for model 3 is the vector of least squares estimates of θ and was made so that the linearization of models would be valid. It implies practically no prejudice because of the large variance chosen.

As a sample calculation we calculate

$$g\{u \text{ given model 3}\}.$$

From the above we have $\theta_0 = \begin{pmatrix} 1.173 \\ .943 \end{pmatrix}$; $V = 100I$; $\sigma^2 = 1$; $n = 4$; $u_i = y_i - 1.173 \exp .943 x_i$

$$\therefore u = \begin{pmatrix} -2.4630 \\ 2.3061 \\ -.6846 \\ .0292 \end{pmatrix}$$

$$X_{ij} = \left[\frac{\partial (\theta_{31} \exp \theta_{32} x_i)}{\partial \theta_{3j}} \right]_{\theta} = \begin{pmatrix} 1.173 \\ .943 \end{pmatrix}$$

$$\therefore X = \begin{bmatrix} 1 & 0 \\ 2.5677 & 3.0119 \\ 6.5930 & 15.4669 \\ 16.9282 & 59.5714 \end{bmatrix}$$

$g\{u \text{ given the model}\} =$

$$\frac{1}{(2\pi)^{n/2} |XVX' + I\sigma^2|^{1/2}} \exp \left\{ -\frac{1}{2} u' [XVX' + I\sigma^2]^{-1} u \right\} \quad (11)$$

This is $N(O, XVX' + I\sigma^2)$, the multivariate normal density function for u with mean O and covariance matrix $XVX' + I\sigma^2$,⁽²⁴⁾

$|A|$ signifies the determinant of any matrix A .

The calculation of the exponent may be simplified by noting that⁽²⁵⁾

$$v'A^{-1}v = \frac{|A + vv'|}{|A|} - 1 \dots \dots \dots (12)$$

On substituting the above values, we get

$$g\{u \text{ given model 3}\} = 4.59 \times 10^{-9}$$

To complete the calculation,

Model	Prior Probability	$g\{u \text{ given the model}\} \times K$	Prior $\times g$	Posterior Probability
1	1/3	0.9939	0.3313	0.082
2	1/3	1.3482	0.4494	0.111
3	1/3	9.8292	3.2764	0.807
			4.0571	

K in the above table is 2.670×10^9 .

It is not the purpose of this exposition to evaluate the two methods used here. They represent two basically different approaches to the problem and when the two methods are applied to choose the "best" model the term "best" has different meanings. Given the definitions of "best" however, either method is acceptable. Likelihood methods are claimed to have an advantage in objectivity in that they "let the data speak for themselves", while the same advantage is claimed for Bayesian methods in that they allow knowledge other than that contained in the data to be used.

Since the particular likelihood method used here compares the models at their best it can be expected that in many cases, as in the example used here, it gives sharper discrimination than the particular Bayesian method used here. However in cases where a reasonable amount of data is available there is little difference.

The design of model discrimination experiments

The basic point of the design of discrimination experiments is to choose conditions which show up differences among the models. While this is obvious it has by no means been sufficiently recognized in some experimental work in the past.

Three different design criteria will be described and they will be illustrated by choosing x_5 , the next value for ξ at which to make a measurement in the example used for the analysis methods.

The Roth criterion⁽²⁰⁾

This design method, like all those to be described, evaluates each possible set of experimental conditions by defining a criterion which measures how much separation between the models would be obtained there. An experimental trial is then made at the conditions which give the best value to the criterion. The experiment is re-analyzed including the new observation and a new set of experimental conditions is chosen by re-evaluating the criterion at each possible set of conditions. This is continued until analysis shows the desired degree of discrimination.

The Roth Criterion calculates a weighted average of the total separation between the models, the weights being the Bayesian posterior probabilities after all the observations which have been made. To choose the experimental conditions for the n th trial it is

$$Z(\xi) = \sum_{i=1}^m \left[P(i, n-1) \prod_{j=1}^m |\hat{y}_{(i)}(\xi) - \hat{y}_{(j)}(\xi)| \right] \dots (13)$$

where ξ is the vector of experimental conditions being evaluated, $P(i, n-1)$ is the posterior Bayesian probability of the i th model after $n-1$ experimental trials and $\hat{y}_{(j)}(\xi)$ is the predicted value of y under experimental conditions ξ using model j with the least squares estimates of the parameters of model j based on the $n-1$ experimental trials. It is calculated below for different values of x_5 in the range $0 \leq x_5 \leq 3$.

The least squares parameter estimates based on the first four trials are $\hat{\theta}_{11} = 5.648$, $\hat{\theta}_{21} = -2.048$, $\hat{\theta}_{22} = 6.526$, $\hat{\theta}_{31} = 1.173$, $\hat{\theta}_{32} = 0.943$. From the Bayesian analysis $P(1, 4) = 0.082$, $P(2, 4) = 0.111$, $P(3, 4) = 0.807$. We have

x_5	$\hat{y}_{(1)}(x_5)$	$\hat{y}_{(2)}(x_5)$	$\hat{y}_{(3)}(x_5)$	$Z(x_5)$
0	0	-2.048	1.173	3.9782
0.5	2.824	1.215	1.880	0.7499
1	5.648	4.478	3.012	3.5618
1.5	8.472	7.741	4.826	9.0319
2	11.296	11.004	7.734	9.5910
2.5	14.120	14.267	12.392	2.6661
3	16.944	17.530	19.857	5.7616

The best separation occurs at $x_5 = 2$ and consequently the next trial should be performed at this value. In this it is obvious from Figure 4 that the best separation occurs about $x_5 = 2$. With more than one independent variable it is difficult to visualize the separation at different points but is still easily calculated.

The Box and Hill criterion^(23, 27)

A deficiency of the Roth criterion is that it fails to take into account what uncertainty there is in the predicted values of the measured variable. It is certainly true that a large predicted separation between two imprecisely known values is not as good as the same predicted separation between two precisely known values. To take this into account Box and Hill have proposed a criterion based on the expected decrease in entropy.

Entropy as used here is a concept introduced to information theory by Shannon⁽²⁸⁻³⁰⁾. It is related to thermodynamic entropy, and it is a measure of the randomness or uncertainty in a situation. If we have m models the i th of which has probability $P(i, n-1)$ after $n-1$ experimental trials we may define the entropy of the system as

$$S = - \sum_{i=1}^m P(i, n-1) \ln P(i, n-1) \dots (14)$$

This is zero when one model is certainly the correct one and it is a maximum when all models have equal probability.

The expected change in entropy resulting from an n th trial made under conditions ξ is shown by Box and Hill⁽²³⁾ to be

$$R = \sum_{i=1}^m P(i, n-1) \int_{y_n} p_i(\xi, y_n) \ln \left[\frac{p_i(\xi, y_n)}{q(\xi, y_n)} \right] dy_n \dots (15)$$

where $p_i(\xi, y_n)$ is the probability density function for y_n , the measurement made at the n th trial assuming model i and experimental conditions ξ . Also

$$q(\xi, y_n) = \sum_{i=1}^m P(i, n-1) p_i(\xi, y_n) \dots (16)$$

Box and Hill show by a Bayesian analysis that if the errors are normally distributed an upper bound for $-R$ is given by

$$D(\xi) = \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m P(i, n-1) P(j, n-1) \left\{ \frac{(\sigma_i^2 - \sigma_j^2)^2}{(\sigma^2 + \sigma_i^2)(\sigma^2 + \sigma_j^2)} + \left[\hat{y}_i(\xi) - \hat{y}_j(\xi) \right]^2 \left(\frac{1}{\sigma^2 + \sigma_i^2} + \frac{1}{\sigma^2 + \sigma_j^2} \right) \right\} \dots (17)$$

where σ^2 is the (known) error variance for a single measurement and $\sigma_i^2 = \mathbf{x}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}'\sigma^2$. $\mathbf{X}$ is as defined previously under Bayesian analysis and $\mathbf{x}$ is the row vector which would be the n th row of $\mathbf{X}$ for an observation made at conditions ξ .

For the example being considered, the following values of D are calculated

x_5	$D(x_5)$
0	0.8385
0.5	0.1113
1	0.5768
1.5	1.4336
2	1.5602
2.5	0.3564
3	0.5259

A method is given in Hill⁽²⁷⁾ for a similar criterion which applies when the error variance is not known in advance.

Expected entropy criterion

It may be argued that an upper bound on expected entropy change should not be used as a design criterion if the actual value of the expected entropy can be calculated.

Following method of calculating the expected entropy has been developed by R. S. Hawkins and the author. The integral in R as defined by Box and Hill⁽²³⁾ can be shown to have the value

$$-\frac{1}{2} [1 + \ln 2\pi (\sigma^2 + \sigma_i^2)] - \int_{y_n} p_i(\xi, y_n) \ln q(\xi, y_n) dy_n$$

The remaining integral can be recognized as the expression for the expectation of $\ln q(\xi, y_n)$ under variation of y_n assuming model i to be correct. This may be written $E_{(i)} [\ln q(\xi, y_n)]$.

It follows that

$$R(\xi) = -\frac{1}{2} (1 + \ln 2\pi) -$$

$$\sum_{i=1}^m P(i, n-1) \left\{ \frac{1}{2} \ln(\sigma^2 + \sigma_i^2) + E_{(i)} [\ln q(\xi, y_n)] \right\} \dots (18)$$

All of this is obviously easy to evaluate except the expectation term. This is the expectation of the function $\ln q(\xi, y_n)$ where y_n is normally distributed with mean $\hat{y}_n(\xi)$ and variance $\sigma^2 + \sigma_i^2$.

The expectation of a function of a normally distributed quantity w with mean m and variance v can be calculated numerically by the following modification of the Gauss-Hermite quadrature

$$E[f(w)] = \frac{1}{\sqrt{2\pi v}} \int_{-\infty}^{\infty} f(w) \exp \left[-\frac{(w-m)^2}{2v} \right] dw \dots (19)$$

$$z = \frac{w-m}{\sqrt{2v}}$$

$$E[f(w)] = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} f(m+z\sqrt{2v}) dz \dots (20)$$

$$E[f(w)] = \frac{1}{\sqrt{\pi}} \sum_{i=1}^n H_i f(m+z_i\sqrt{2v}) \dots (21)$$

where the z_i are the zeros of the n th degree Hermite polynomial and H_i are the weights for the n th order Gauss-Hermite quadrature⁽³¹⁾.

It is easy to use Equation (21) for calculating the expectation required for R_i .

This was done using fourth order quadrature for the numerical example used previously with the results

x_b	$-R(x_b)$
0	0.2606
0.5	0.0476
1	0.1961
1.5	0.3371
2	0.3429
2.5	0.1431
3	0.2138

This criterion, by the definition of the Box and Hill criterion, always gives a higher value than the Box-Hill one. The difference is not constant and the true expected entropy criterion is worthy of investigation, at least for the purpose of assessing in what situations, if any, the Box-Hill criterion fails to choose the best design.

The results for the three criteria which were used on the numerical example are summarized below.

x_b	Roth	Box & Hill	Expected Entropy Decrease
0	3.9782	0.8385	0.2606
0.5	0.7499	0.1113	0.0476
1	3.5618	0.5768	0.1961
1.5	9.0319	1.4336	0.3371
2	9.5910	1.5602	0.3429
2.5	2.6661	0.3564	0.1431
3	5.7616	0.5259	0.2138

Note that $x_b=2$ was chosen in every case.

General comments

The methods which were presented here were given as examples of the types of approaches which can be used. No attempt has been made to show or even to mention all the methods which are available. This particular field is one which now is developing rapidly but it can be expected that methods will become more standardized in the future. It is hoped to publish in the near future some practical results to which the likelihood method described in this paper has been applied.

References

The references listed below have been quoted in the text with the exception of Box and Hunter (1965). This is an excellent paper on the general philosophy of the experimental study of physical mechanisms. Together this selection of papers constitutes a reasonable background in the subject.

- (1) Box, G. E. P., *Biometrics*, **10**, 16 (1954).
- (2) Box, G. E. P., *Bull. Inst. Intern. De Statistique*, **36**, 215 (1958).
- (3) Box, G. E. P., *Annals of the New York Academy of Sciences*, **86**, 792 (1960).
- (4) Box, G. E. P. and Draper, N. R., *J. Am. Stat. Assoc.*, **54**, 622 (1959).
- (5) Box, G. E. P. and Draper, N. R., *Biometrika*, **50**, 335 (1963).
- (6) Box, G. E. P. and Hunter, J. S., *Ann. Math. Stat.*, **28**, 195 (1957).
- (7) Box, G. E. P. and Hunter, J. S., *Technometrics*, **3**, 311 and 449 (1961).
- (8) Box, G. E. P. and Hunter, W. J., *Technometrics*, **7**, 23 (1965).
- (9) Box, G. E. P. and Youle, P. V., *Biometrics*, **11**, 287 (1955).
- (10) Box, G. E. P. and Wilson, K. B., "On the Experimental Attainment of Optimum Conditions", *J. R. Stat. Soc. (B)*, **13**, 1 (1951).
- (11) Hill, W. J. and Hunter, W. G., *Technometrics*, **8**, 571 (1966).
- (12) Davies, O. L. (ed.), "The Design and Analysis of Industrial Experiments", Oliver and Boyd (1956).
- (13) Hogen, O. A. and Watson, K. M., "Chemical Process Principles", Part Three, "Kinetics and Catalysis", Wiley.
- (14) Kittrell, Mezaki and Watson, "Model Building Techniques for Heterogeneous Kinetics", *British Chemical Engineering* (1966).
- (15) Cox, D. R., *Proc. Fourth Berkeley Symp.*, **1**, 105 (1961).
- (16) Cox, D. R., *J. R. Stat. Soc. (B)*, **24**, 406 (1962).
- (17) Barnard, G. A., Jenkins, G. M. and Winsten, C. B., *J.R. Stat. Soc. (A)*, **125**, 321 (1962).
- (18) Jenkins, G. M. and Watts, D. G., "Spectral Analysis", Holden Day (1968).
- (19) Sprott, D. A. and Kalbfleisch, J. D., 1969, "Marginal and Conditional Likelihood", To be published.
- (20) Lindley, D. V., "Introduction to Probability and Statistics from a Bayesian Viewpoint", Cambridge (1965).
- (21) Jeffreys, H., "The Theory of Probability", Clarendon Press, Oxford (1961).
- (22) Reilly, P. M. and Johri, H. P., *Chemical Engineering*, **122**, Apr. 7, 1969.
- (23) Box, G. E. P. and Hill, W. J., *Technometrics*, **9**, 57 (1967).
- (24) Plackett, R. L., "Principles of Regression Analysis", Oxford (1960).
- (25) Scheffé, H., "The Analysis of Variance", Wiley, p. 417 (1959).
- (26) Roth, P. M., "Design of Experiments for Discrimination Among Rival Models", Ph.D. Thesis, Princeton University (1965).
- (27) Hill, W. J., "Statistical Techniques in Model Buildings", University of Wisconsin Ph.D. Thesis (1966).
- (28) Shannon, C. E., *Bell System Tech. Journal*, **27**, 379 and 623 (1948).
- (29) Khinchin, A. Y., "Mathematical Foundations of Information Theory", Dover (1957).
- (30) Pierce, J., "Symbols, Signals and Noise", Hutchison (1962).
- (31) Lapidus, Leon, "Digital Computation for Chemical Engineers", McGraw-Hill (1962).

Manuscript received November 17, 1969; accepted March 11, 1970.

★ ★ ★